

Local Python Setup Guidelines for Agentic AI

This programme involves hands-on Python development for building and evaluating agentic systems. A hosted environment (e.g., Vocareum) will be available throughout; however, such platforms may experience latency or kernel interruptions during periods of heavy usage. Configuring a Python development environment on your workstation provides:

- Reliable, offline access without service interruptions
- Capacity to execute larger or more complex workloads efficiently
- Professional practice in installing, managing, and troubleshooting a development toolchain

What You'll Need

You'll set up three components:

1. **Python** (version 3.10 or higher)
2. An IDE (Integrated Development Environment) – choose VS Code or PyCharm
3. **Essential Libraries** (installed via pip)

Step-by-Step Setup Instructions

Step 1: Install Python

1. Go to <https://www.python.org/downloads/>.
2. Download the latest **Python 3.x** version for your operating system (Windows/Mac/Linux).
3. During installation:
 - Check the box "Add Python to PATH".
 - Then click **Install Now**.
4. Once installed, open a **terminal/command prompt** and verify:
5. `python --version`

It should show something like Python 3.11.6.

Step 2: Install an IDE (Choose One)

Option 1: Visual Studio Code (Recommended)

1. Download from <https://code.visualstudio.com/>.
2. Open VS Code → Go to Extensions (left toolbar icon).
3. Install the following:
 - **Python** (Microsoft)
 - **Jupyter** (for running notebooks)

- **Pylance** (for better IntelliSense)

Option 2: PyCharm (Community Edition)

1. Download from <https://www.jetbrains.com/pycharm/download/>.
2. Choose the **Community Edition** (it's free).
3. Follow the on-screen setup wizard to link Python automatically.

Step 3: Create a Project Folder

- Make a new folder on your computer, for example:
- C:\AI_Projects
- Open it in your IDE.
- Inside, create a test file:
- `print("Hello, Agentic AI!")`
- Run it. If it prints correctly, your setup is working.

Step 4: Install Core Libraries

In your terminal (inside the IDE), run:

```
pip install numpy pandas matplotlib langchain openai crewai
```

If you see messages like “Requirement already satisfied,” that’s normal — it means it’s already installed.

Step 5: Check Jupyter Notebook Support

To run notebooks locally:

```
pip install jupyterlab
```

Then open it with:

```
jupyter lab
```

You can create .ipynb notebooks just like in Vocareum.

Troubleshooting Common Issues

Problem	Possible Fix
python not recognized	Reinstall Python and ensure “Add to PATH” is checked.
Slow or stuck pip install	Use a faster mirror: <code>pip install -i https://pypi.org/simple package_name</code>
Notebook doesn't open	Ensure Jupyter is installed and run <code>jupyter notebook</code> again.
Library import error	Reinstall the library using <code>pip install --upgrade package_name</code> .

Important Notes

- You will continue using **Vocareum** for graded exercises.
- The local setup is for practice, debugging, and exploration — especially during live coding weeks.
- For further assistance, please get in touch with your Programme Leader.